Basic Hydrologic Concepts

Katie Markovich, EPS 522/ ENVS 423L (Fall 2025)

Announcements

- Lots of changes to the repo, pull now if you haven't yet.
- Canvas site is live!
- No lab today, first lab will be next Thursday, August 28.
- First problem set will be posted to Canvas after class.
- Course textbook

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- Fick's First Law of Diffusion, a substance moves from higher to lower concentrations at a rate that is proportional to the concentration gradient.

Primary Dimensions

A dimension is the type of physical quantity and a unit is a means of assigning a numerical value to that quantity.
Science uses international system of units (SI), but practitioners in hydrology often still use imperial units.

1.
2.
3.
4.
5.
6.
7.
8.

Derived Quantities

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5. Density, ρ, (M/V)

Dimensional Analysis

Analysis of relationships between different physical quantities by identifying their base dimensions and units. A fancy way of saying, check the units!

What are the dimensions of Force and Pressure?

State Variables

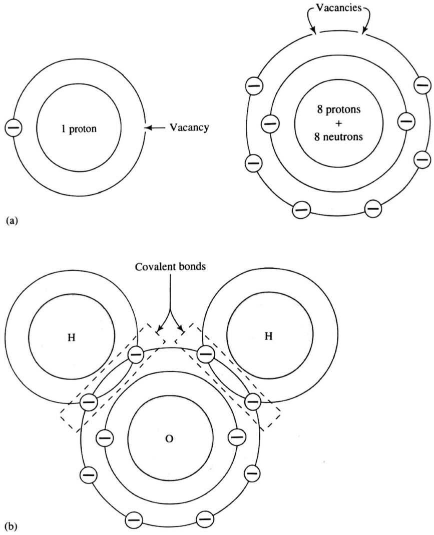
State variables describe a condition of a system at a specific time and location. Examples include:

-
-
-
-
-
-
-

Water Molecule

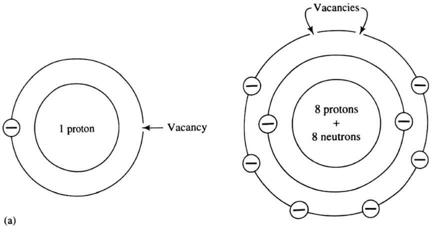
The molecular formula of water is H_2O, with two hydrogen atoms and one oxygen atom. The outer shell of hydrogen has 1 electron but can accommodate 2 and the outer shell of oxygen has 6 electrons but can accommodate 8.

This mutual sharing of electrons is a **covalent bond**.

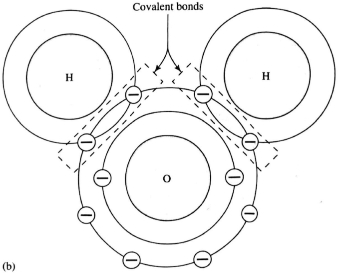


Features of Water Molecule

- Covalent bonds are very strong (much energy needed to break them)



(a)



(b)

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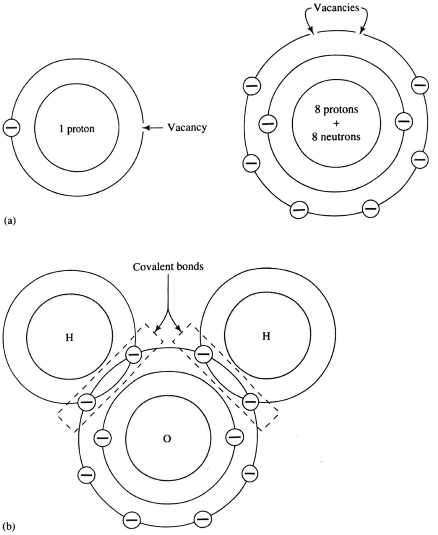

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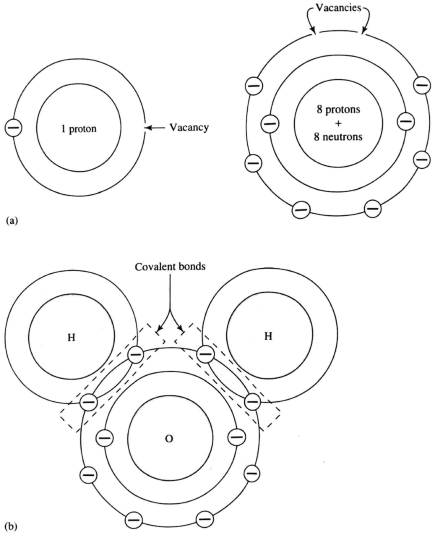

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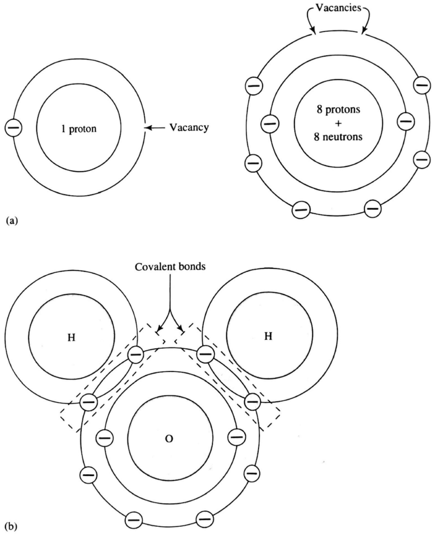

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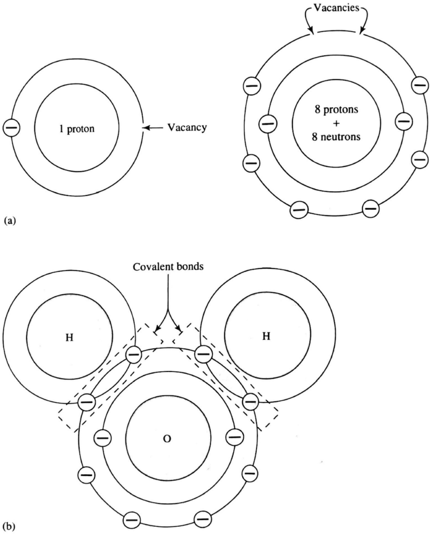
- These features explain much of the unusual properties of water.



Thermal Capacity

The strong hydrogen bonds of the water molecule lead to a high thermal capacity, or the energy (ΔH in Joules) required to raise one kilogram of water 1 K

$$c_p = \frac{\Delta H}{M \cdot \Delta T} \tag{1}$$

At standard temperature and pressure c_p = 4216 J kg^{-1} K^{-1}, and this value decreases slightly with increasing temperature.

Thermal Capacity

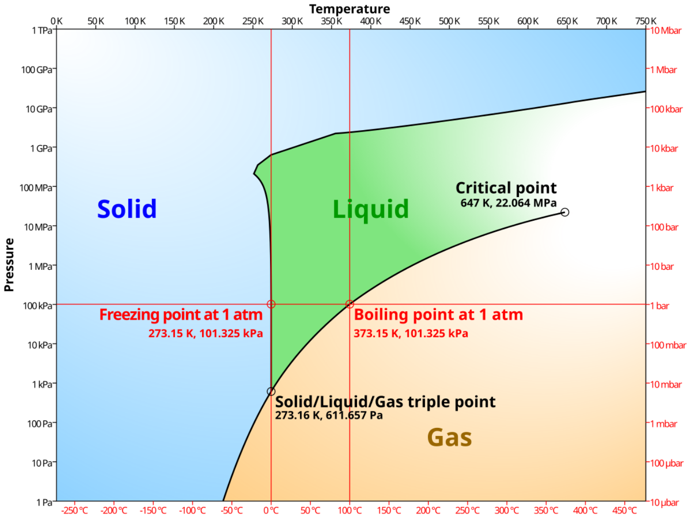
This relatively high heat capacity is critical for warm-blooded organisms to regulate their body temperature and for oceans/bodies of water to moderate the rates of ambient temperature changes.

Substance	c_p (J/kg°K)*	ρ (kg/m^3)
Freshwater (liquid)	4.186×10^3	1.00×10^3
Freshwater (ice)	2.108×10^3 (at 0 °C)	0.92×10^3
Seawater	3.93×10^3 (at 2.2 °C)	1.025×10^3
Air	1.005×10^3	1.205 (at 20 °C)
Granite	7.9×10^2	**2.75×10^3
Marble	8.8×10^2	**2.80×10^3
Sandstone	9.2×10^2	**2.80×10^3
Limestone	8.4×10^2	**2.70×10^3
Dolostone	9.2×10^2	**2.90×10^3
Clay	9.2×10^2	*$1.8 \times 10^3 - 2.6 \times 10^3$
Quartz	7.1×10^2 (at 0 °C)	**2.64×10^3
Calcite	8.4×10^2 (average over 0–100 °C)	**2.71×10^3

*Engineering Tool Box (2007).

**Cote and Konrad (2005).

Phase Diagram



Density

Mass density (ρ) is mass per unit volume (M/L^3)

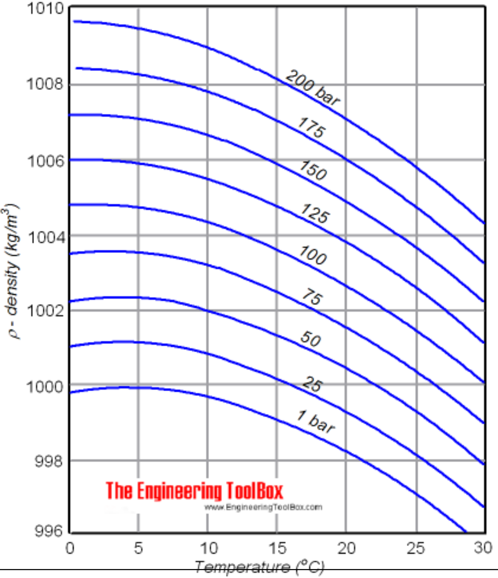
$$\rho = \frac{M}{V}$$

Weight density (γ) is weight per unit volume,

where weight is gravitational force F, resulting in dimensions of (F/L^3). These two are related to each other via Newton's Second Law.

$$\gamma = \rho \cdot g$$

where g is gravitational acceleration (9.81 m/s^2)



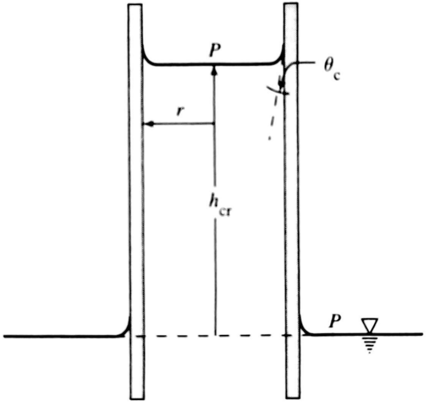
Surface Tension

Hydrogen bonds produce a net inward force to molecules at the surface of water called surface tension (σ) having units of force (F) divided by distance over which it acts (L^{-1})

σ is equal to 0.0756 Nm^{-1} at $0^{\circ}C$ but decreases with increasing temperature or in presence of dissolved surfactants.

Capillary Rise

The polarity of water also leads to an attraction of molecules to surfaces, leading to water being drawn upward where flow scale is less than a few millimeters (i.e., pores).



$$F_u = \sigma \cdot \cos(\theta_c) \cdot 2 \cdot \pi \cdot r$$

$$F_d = \gamma \cdot \pi \cdot r^2 \cdot h_{cr}$$

Where, σ is surface tension, θ_c is contact

angle, r is radius of cylinder, γ is weight density of water, and h_{cr} is height to which water will rise.

Solving for Capillary Rise

Since Newton's 3rd Law of Motion states that for every action force, there is an equal and opposite reaction force, we can combine the equations of upward and downward force to solve for capillary rise (h_{cr}).

We know (or can estimate) σ and γ, and we can measure θ_c and r, we can solve for the expected capillary rise in a cylinder.

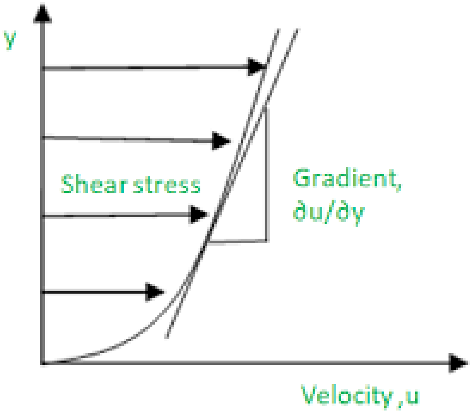
Viscosity

Viscosity, generally, is resistance to flow. For water, this is due to the hydrogen bonds, where water molecules tend to "stick" to a boundary as they flow past it.

Dynamic viscosity (μ) from Newton's Law of Viscosity is:

$$\tau = \mu \cdot \frac{du}{dy}$$

where τ is shear stress ($\frac{F}{A}$) and $\frac{du}{dy}$ is the rate of change of velocity.



Kinematic viscosity (*nu*) is

$$\nu = \frac{\mu}{\rho}$$

Standard Temperature and Pressure (STP)

The properties of liquids and gases change with changes in temperature and pressure, so we use STP as a reference point.

Standard temperature is 273.15 K (or 0 degrees Celsius and 32 degrees Fahrenheit) and standard pressure is 1 atmosphere (or 1 bar, 101.325 kPa, and 14.696 psi), which is the standard air pressure at sea level.

Temperature Dependence of Properties

Temperature (°C)	ρ, γ	μ	ν	σ	c_p	λ_v
0	1.00000	1.0000	1.0000	1.0000	1.0000	1.0000
3.98	1.00013					
5	1.00012	0.8500	0.8500	0.9907	0.9963	0.9953
10	0.99986	0.7314	0.7315	0.9815	0.9940	0.9904
15	0.99926	0.6374	0.6379	0.9722	0.9924	0.9857
20	0.99836	0.5607	0.5616	0.9630	0.9915	0.9810
25	0.99720	0.4983	0.4997	0.9524	0.9910	0.9763
30	0.99580	0.4463	0.4482	0.9418	0.9907	0.9715

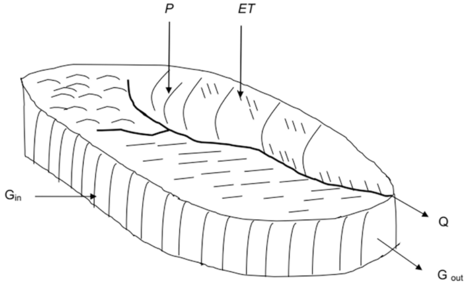
$\rho \equiv$ mass density, $\gamma \equiv$ weight density, $\mu \equiv$ dynamic viscosity, $\nu \equiv$ kinematic viscosity, $\sigma \equiv$ surface tension, $\lambda_v \equiv$ latent heat of vaporization, $c_p \equiv$ specific heat.

Conservation Equations

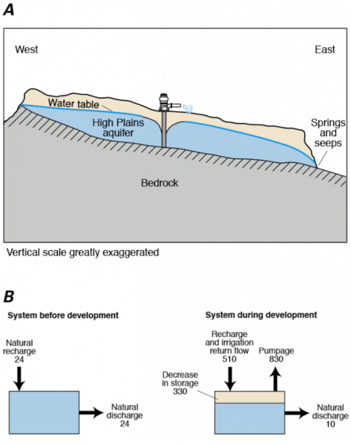
Amount In - Amount Out = Change in Storage

for a defined control volume, a defined period of time, and a conservative substance.

Control Volume



Where:
P = Precipitation
ET = Evapotranspiration
Q = Stream outflow
G_{in} = Ground water in
G_{out} = Ground water out



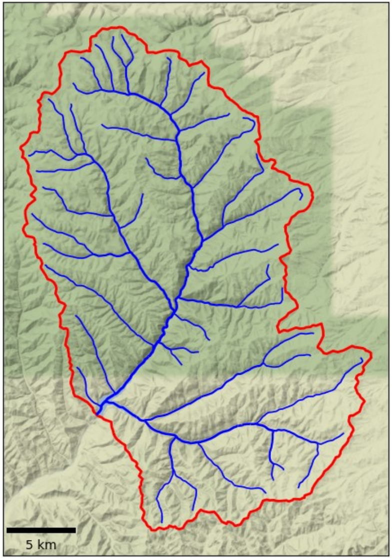
Does not have to be physically contiguous (i.e., glaciers)

Watershed Delineation

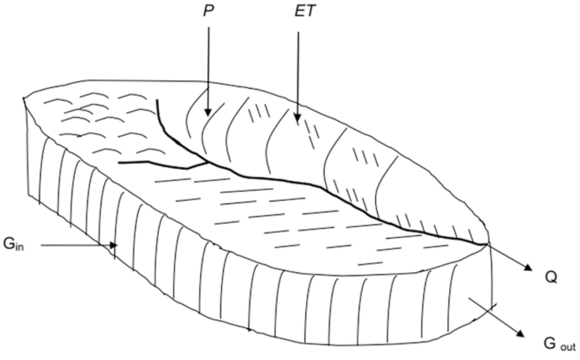
Watershed is most common control volume, defined as area that appears (on basis of topography) to contribute all water that passes through a given cross-section of a stream.

Nowadays, it is trivial to delineate a watershed based on an arbitrary point along a stream using a Digital Elevation Model (DEM) and openly available GIS tools.

`https://mghydro.com/watersheds/`



Regional Water Balance



$$P + G_{in} - (Q + ET + G_{out}) = \Delta S$$

Where:

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Challenges to Estimating Regional Water Balance

Hydrologic Modeling

We use models to predict or forecast how changes in climate and land use will impact the water balance, groundwater availability, water quality, etc.

What is a model?

A representation of a portion of the real world "which is simpler than the prototype system and which can reproduce some but not all of the characteristics thereof" (Dooge, 1986).

Types of Models

- **Physical model**: tangible reconstructed portion of the real-world.
  Example
- **Analog model**: Use physically analogous process to simulate a natural process.
  Example
- **Mathematical model**: Use equations (can be exact or approximate solutions) to represent a real-world system. These equations include **parameters**, which are constants that dictate hydrologic behavior.
  Example

EPS 522 Term Project

Hydrologic Analysis of a Control Volume

1. Find a partner (should be 8 groups of 2)

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4. Assemble all available data and quantify the regional water balance.
5. Identify the primary water resource management concern(s) (e.g., climate change, land use change, contamination, drought, etc.) and outline potential next steps.

EPS 522 Term Project

Hydrologic Analysis of a Control Volume

Deliverables: technical memorandum (due December 2) and presentation to the class (week of December 2).

Intermediate Deadlines: Submit a brief proposal with the group member names and control volume (September 2), submit the literature review (October 2)

Problem Set #1

- Online submission (docx/pdf)
- Work together, use your time with the TA in next week's lab
- Use whatever tools you want (python/Excel/calculator/chatGPT)
- Show your work! List assumptions, assumed values for variables, intermediate solutions, etc.